A Mechanistic Study of AI-Detection Features in Frozen BERT: Sparse Probing and Activation Patching on RAID

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Abstract

AI-generated text detectors achieve high accuracy on standard benchmarks, yet the internal representations that drive these predictions remain poorly understood. We study which neurons in a *frozen*, unmodified BERT-base-uncased encoder support AI-text detection, using the RAID benchmark across six generators spanning pure-base and instruction-tuned architectures. Applying the L1→L2 sparse-probing protocol of Gurnee et al. (2023) to all 9,216 CLS-token features (12 layers × 768 dimensions), we identify stable sets of 45–62 neurons per generator (<1% of features) that achieve 90.5–98.8% detection accuracy with high cross-fold reproducibility (Jaccard 0.67–0.76). To assess the causal role of these neurons, we apply same-domain activation patching: transplanting the selected neurons' activations from AI donors into human targets flips 1.07–8.15% of predictions to AI — 10–23× above a random-neuron baseline — while mean-ablating the same set leaves accuracy largely intact (≤1.1 pp drop on 5/6 generators), suggesting the signal is redundantly distributed. Cross-generator analysis reveals a bipartite structure: instruction-tuned generators concentrate 30–36% of stable neurons in BERT's final layer, while both base generators fall below 14%, consistent with a layer-12 footprint of post-training alignment. Leave-one-family-out evaluation shows that the selected neurons retain 86–94% of the full-feature ceiling on entirely unseen generator families.

1 Introduction

The rapid development of Large Language Models (LLMs) has made AI-generated text increasingly difficult to distinguish from human writing (Dugan et al., 2024). This poses a significant challenge across domains ranging from academic integrity to social media moderation and journalist authenticity. Although detection accuracy has improved substantially, with encoder-based classifiers reaching high performance on established benchmarks (Dugan et al., 2024), a fundamental question remains unanswered: *What internal representations drive these predictions?* Mechanistic interpretability has seen little application to AI-text detection, leaving the field without a causal account of how detection actually works.

Existing interpretability work on AI-text detection takes one of three forms: a) *black-box feature attribution*, which identifies input-level correlates via SHAP or LIME without inspecting model internals (Pudasaini et al., 2026); b) *confounder removal*, which suppresses neurons correlated with domain or style artifacts in fine-tuned encoders to improve out-of-distribution accuracy (Borile and Abrate, 2025); or (c) *sparse autoencoder feature analysis* applied to decoder-based models (Kuznetsov et al., 2025). These approaches share a fundamental limitation: they are *observational*. They identify neurons that correlate with detector behavior or remove features that hurt generalization, but none directly tests whether a small set of internal representations is *causally sufficient* to produce AI-text predictions.

We address this gap with a causal intervention study on a *frozen* BERT-base-uncased (Figure 1). Rather than fine-tuning the model for detection, we train sparse linear probes on pretrained CLS-token activations concatenated across all 12 layers (9,216 features total), following the sparse-probing protocol of Gurnee et al. (2023). An L1-regularized logistic regression selects a small stable set of candidate neurons; an independent L2 probe serves as the frozen evaluator for all subsequent experiments. We then apply two causal interventions to the same selected feature set: *mean ablation*, which replaces these features with their training-set means to test whether they are necessary, and *same-domain donor → target activation patching*, which replaces their activations in human samples with those from AI donors to test whether they are

sufficient to flip detector predictions. The patching direction is the key distinction from prior work: rather than removing suspected confounders, we inject candidate detection signal and measure the result.

Our experiments suggest that such sufficiency holds. Across six generators spanning two pure-base and four instruction-tuned model families, patching $\sim$50 selected neurons flips human $\rightarrow$ AI predictions at 1.07–8.15%, 10–23$\times$ above a random-feature baseline — providing causal-sufficiency evidence for AI-detection circuits in a frozen pretrained encoder. Mean-ablation of the same features causes only 0.03–1.06 pp accuracy loss, suggesting that the detection signal is *redundantly distributed*: easy to inject, hard to erase. Cross-generator analysis shows that stable feature sets are largely generator-specific (pairwise Jaccard 0.00–0.24) yet cluster $\sim$24$\times$ above chance, with instruction-tuned generators concentrating 30–36% of stable neurons in layer 12 while both pure-base generators (GPT-2, MPT-7B) have $\leq$14%.

Our work is most directly related to Borile and Abrate (2025), who suppress confounding neurons in a fine-tuned encoder to improve out-of-distribution detection, an observational approach that tests necessity-for-confounding rather than causal sufficiency. We discuss the relationship to this and other prior work in Section 2.

Contributions.

1. A sparse-probing characterization of AI-detection signal in frozen BERT-base-uncased across six generators on RAID, identifying stable sets of 45–62 neurons ($<$1% of 9,216 features) sufficient for 90.5–98.8% detection accuracy with high cross-fold stability (Jaccard 0.67–0.76), with hyperparameters validated via a multi-generator stability sweep.

2. The first causal-sufficiency test for AI-detection features via same-domain donor $\rightarrow$ target activation patching, yielding flip rates 10–16$\times$ above a random-feature baseline across all six generators.

3. A cross-generator circuit analysis revealing generator-specific stable sets that nonetheless overlap $\sim$24$\times$ above chance, with instruction-tuned generators concentrating 30–36% of stable neurons in layer 12 versus only 4–13% for the two base generators (GPT-2, MPT-7B), supported by a base-model cluster Jaccard of 40$\times$ chance.

4. Leave-one-family-out (LOGO) generalisation showing that $\leq$1.4% of neurons (91–125 out of 9,216) retain 86–94% of the full-feature detection signal on five unseen model families (Sel L2 accuracy 76.3–93.7%).

2 Related Work

2.1 AI-Generated Text Detection

Supervised detectors fine-tune pretrained language models as binary classifiers on pairs of human and machine-generated text. The GPT-2 output detector (Solaiman et al., 2019) established this paradigm by fine-tuning RoBERTa on GPT-2 samples; subsequent work extended the approach to instruction-tuned and RLHF-aligned generators with larger and more diverse training sets. These classifiers achieve high in-distribution accuracy but degrade substantially on generators or domains unseen during training (Dugan et al., 2024).

Zero-shot methods exploit statistical properties of language model distributions without requiring labelled data. DetectGPT (Mitchell et al., 2023) tests whether candidate text lies at a local maximum of log-probability relative to nearby perturbations, reflecting the tendency of model-generated text to occupy high-probability regions. Binoculars (Hans et al., 2024) uses the cross-perplexity ratio between two models as a calibrated detection score, achieving strong performance across generators. Both approaches require whitebox or API access to a reference model.

Systematic benchmarking has revealed that no single method generalises reliably. RAID dataset (Dugan et al., 2024) spans 11 generators and 6 domains with adversarial rephrasing attacks; classifiers that perform well in-distribution consistently fail on held-out generator families or domains. This pattern motivates our focus: rather than improving detection accuracy, we ask what internal representations actually drive it.

2.2 Interpretability of Text Detectors

The shallowest form of interpretability for AI-text detectors uses input-level attribution. Pudasaini et al. (2026) apply SHAP to explain classifier decisions, finding that the most predictive tokens diverge substantially across domains. This approach characterises the model's decision surface but cannot speak to what internal features underlie it.

More recent work examines representations inside the encoder. Borile and Abrate (2025) identify neurons in a fine-tuned BERT encoder whose activations correlate with domain or style artefacts rather than AI-authorship signal, and suppress them to improve out-of-distribution accuracy. Kuznetsov et al. (2024) take a geometric approach, removing a learned embedding subspace to produce a detector more robust to lexical variation. Both works operate on fine-tuned models and test necessity-for-confounding: the question is which features hurt generalisation, not which features are sufficient to drive predictions. Our work differs from Borile and Abrate (2025) on four axes: we use a *frozen* rather than fine-tuned encoder; we probe full-layer concatenated CLS representations rather than intermediate FFN activations; we apply activation patching to test *sufficiency* rather than ablation to identify confounders; and we evaluate on RAID rather than DAIGT/M4/HC3.

Kuznetsov et al. (2025) apply sparse autoencoders to Gemma-2 decoder activations, extracting semantically interpretable features correlated with AI authorship. This work is complementary but distinct: the model class (autoregressive decoder), representation type (token-position activations rather than CLS pooling), and analysis method (SAE reconstruction) all differ from ours, and the approach remains observational – features are identified by correlation, not by causal intervention. None of the above provides direct evidence that a small identified set of neurons is *sufficient* to produce AI-text predictions when injected into human-text activations; that is the question we address.

2.3 Sparse Probing and Activation Patching

Sparse probing uses L1-regularised linear classifiers to identify a small set of neurons in a pretrained model that carry task-relevant information. Gurnee et al. (2023) introduced and systematically evaluated the L1-then-L2 protocol: an L1 probe selects candidate neurons, and a re-fitted L2 probe on those neurons provides unbiased accuracy estimates. Their study demonstrates that individual neurons in GPT-2 and LLaMA encode grammatical, factual, and linguistic properties, and that the selected sets are stable across seeds and data subsets. We adopt this protocol directly, applying it to BERT-base-uncased CLS-token activations concatenated across all 12 layers.

Activation patching replaces internal representations in a target forward pass with those from a donor pass, then measures the effect on model outputs. Vig et al. (2020) introduced causal mediation analysis for language models, using interventions on attention heads to trace gender bias; Meng et al. (2022) applied patching at MLP layers in GPT to localise factual associations. Heimersheim and Nanda (2024) provide a systematic treatment of patching variants and recommend donor patching – replacing with activations from a real input rather than a zero or mean vector – to avoid out-of-distribution artefacts, and mean ablation as a conservative necessity baseline. We follow both recommendations: selected neurons are mean-ablated for the necessity test and replaced with activations from same-domain AI donors for the sufficiency test.

The closest methodological precedent for our flip-rate framing is Ravindran (2025), who apply donor-style activation patching to test whether deceptive activations from an aligned model can cause unsafe outputs in a target pass. The setting differs substantially – autoregressive decoder, safety alignment rather than AI-text detection – but the core intervention and the use of prediction-flip rate as the primary metric are shared. To our knowledge, no prior work applies this methodology to encoder-based AI-text detection.

3 Setup

3.1 Data: RAID

RAID (Dugan et al., 2024) provides human-written and machine-generated texts across 11 generators and 8 domains. We use six of the eight domains: abstracts, books, news, Reddit posts, reviews, and Wikipedia. We select six generators for architectural diversity: GPT-4 (`gpt4`), GPT-2 (`gpt2`), MPT-7B (`mpt`), Mistral-7B-Instruct-v0.1 (Jiang et al., 2023) (`mistral-chat`), Llama-2-Chat (`llama-chat`), and Cohere Command (`cohere-chat`), spanning two pure-base generators (GPT-2, MPT-7B), one SFT-only generator (Mistral-7B-Instruct-v0.1, which Jiang et al. describe as fine-tuned on publicly available instruction datasets with no RLHF or preference optimisation), and three SFT+RLHF/preference-trained generators (GPT-4, LLaMA-2-Chat, Cohere Command), across five model families. We subsample 7,500 examples per generator (3,750 human, 3,750 AI) balanced across the six RAID domains, yielding 45,000 samples in total.

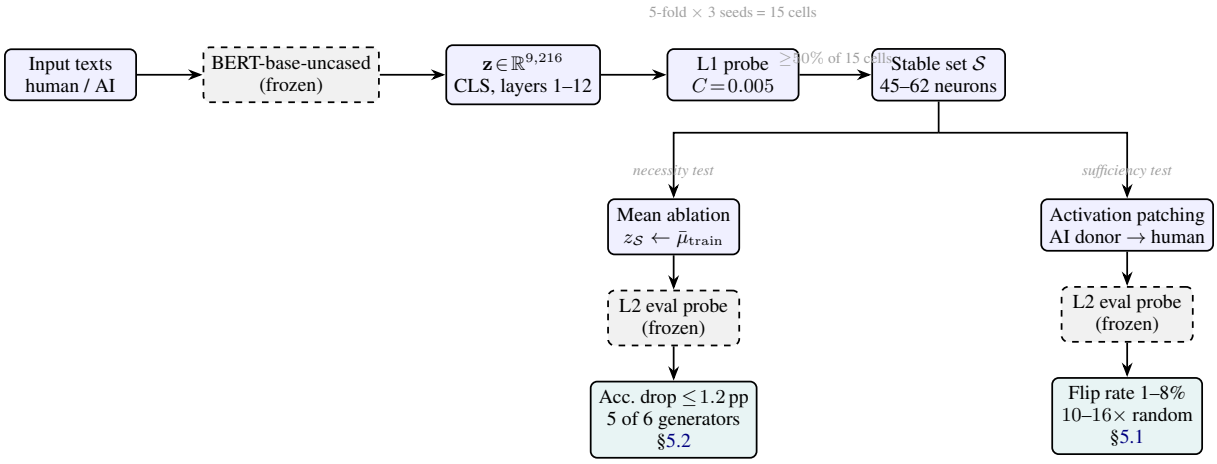

Figure 1: Overview of the experimental pipeline. Frozen BERT-base-uncased CLS activations are concatenated across all 12 layers, yielding $\mathbf{z} \in \mathbb{R}^{9,216}$. An L1-regularised probe ($C = 0.005$) selects a sparse candidate set per (fold, seed) cell; neurons stable in $\geq 50\%$ of all 15 cells become the *stable set* $\mathcal{S}$ (45–62 neurons per generator). An independent L2 probe trained on all 9,216 features is frozen and reused as the decision function in both experiments. **Left branch** (§5.2): $\mathcal{S}$ activations are replaced with their training mean (*mean ablation*), testing *necessity* — accuracy drops by ≤ 1.2 pp on 5 of 6 generators, indicating the signal is redundantly distributed. **Right branch** (§5.1): $\mathcal{S}$ activations are transplanted from a same-domain AI donor into a human sample (*activation patching*), testing *causal sufficiency* — 1–8% of human samples flip to AI predictions, 10–16× above a random-neuron baseline.

3.2 Model and Activations

We use BERT-base-uncased (Devlin et al., 2019) with all weights frozen throughout. For each input text we extract the CLS-token hidden state at every transformer layer $l \in \{1, \ldots, 12\}$, giving a per-layer vector $\mathbf{h}^{(l)} \in \mathbb{R}^{768}$. The full feature representation is the concatenation

$$\mathbf{z} = \left[\mathbf{h}^{(1)}; \ldots; \mathbf{h}^{(12)}\right] \in \mathbb{R}^{9,216}.$$

We refer to individual dimensions of $\mathbf{z}$ as *neurons*; neuron index i resides in layer $\lfloor i/768 \rfloor + 1$ and position $i \bmod 768$ within that layer. Using the frozen pretrained model (rather than a fine-tuned detector) ensures that any informative features we identify are present in general-purpose BERT representations, not artefacts of task-specific fine-tuning.

3.3 Protocol: L1 Selection, L2 Evaluation

Following Gurnee et al. (2023), for each (seed, fold) cell we fit two independent probes on the training split. (1) *L1 selection probe*: L1-regularised logistic regression ($C = 0.005$, liblinear solver) on all 9,216 features. Neurons with non-zero coefficients form the *candidate set* $\mathcal{S}$; neurons appearing in $\mathcal{S}$ in at least 50% of the 15 cells are *stable neurons*. (2) *L2 evaluation probe*: L2-regularised logistic regression (lbfgs) on the full $\mathbf{z}$. This probe is held frozen for all downstream experiments to ensure that intervention effects are measured against a fixed decision function.

We use 5-fold stratified cross-validation stratified by (label × domain) with seeds $\{42, 123, 456\}$, yielding 15 cells per (generator, experiment). All reported metrics are 15-cell means ± standard deviation unless otherwise noted.

3.4 Stability-Driven Hyperparameter Selection

The regularisation strength $C = 0.005$ and sample size $N = 7{,}500$ were selected via a multi-generator stability sweep conducted prior to any results analysis, following the criterion of Gurnee et al. (2023). We drew $K = 15$ stratified subsamples from each of three generators (`gpt4`, `llama-chat`, `mistral-chat`), computed mean pairwise Jaccard similarity between selection sets across subsamples, and chose the (C, N) pair that maximised the minimum Jaccard across generators. At $C = 0.005$, $N = 7{,}500$, the minimum-over-generators Jaccard is 0.724 — the highest achieved by any C at every N tested. The full sweep is reported in Appendix B.

4 Sparse Detection Circuits in Frozen BERT

4.1 Stable Neuron Sets

Table 1 reports sparse-probe results across all six generators. Between 45 and 62 neurons — 0.5–0.9% of the 9,216-feature CLS representation — are stable across all 15 cells, yet the L2 evaluation probe achieves 90.5–98.8% validation accuracy (AUC-ROC 0.968–0.999). Selection is highly reproducible: pairwise fold Jaccard ranges 0.67–0.76 across 105 fold-pairs per generator, compared to a random-null expectation of ∼0.003–0.005 for sets of this size drawn from 9,216 features, placing within-generator stability at 130–200× above chance. The selection pool is narrow: between 91 neurons ever selected (llama-chat) and 146 (gpt2), indicating that the probe consistently returns to a small region of the feature space.

Detection difficulty varies systematically across generators. GPT-4 is easiest to detect (98.8% val acc, AUC 0.999), followed by LLaMA-2-Chat (98.6%) and Mistral-7B-Instruct (97.1%); all three are instruction-tuned and produce outputs that diverge markedly from human writing in BERT's frozen representation space. MPT-7B (97.2%) is close behind despite being a pure-base generator, with 52 stable neurons forming a compact and highly reproducible selection set. GPT-2 sits at the mid-point (95.9%) with the loosest fold Jaccard (0.670) and largest observed pool (146 neurons) among all six generators, suggesting a noisier but stable feature landscape for this base model. Cohere-Command-Chat is the structural difficulty floor at 90.5% val accuracy and AUC 0.968, with the highest standard deviation on n_{sel} (CV 4.3%), consistent with its smaller per-axis signal magnitude documented in Section 6. Where in the network these neurons reside — and the striking asymmetry between base and instruction-tuned generators — is examined in Section 6.

4.2 Detection Signal Geometry

To characterise the geometry of the detection signal within each stable set we compute a concept activation vector (CAV) as the mean difference between AI and human activations restricted to the stable neurons, and compare it to the L1 probe weight vector. Table 2 reports cosine similarity between the CAV and the probe weight, logistic-regression accuracy on the CAV projection, and the norms of both vectors.

Generator	Val acc	AUC	n_{sel}	Jaccard	n_{stable} / n_{obs}
GPT-4	0.988 ± 0.003	0.999 ± 0.000	74.1 ± 2.9	0.763 ± 0.037	60 / 106
GPT-2	0.959 ± 0.003	0.992 ± 0.001	84.7 ± 4.1	0.670 ± 0.042	60 / 146
MPT	0.972 ± 0.003	0.994 ± 0.001	61.3 ± 3.1	0.740 ± 0.036	52 / 94
Mistral	0.971 ± 0.004	0.995 ± 0.001	57.8 ± 2.9	0.724 ± 0.046	45 / 99
LLaMA	0.986 ± 0.003	0.998 ± 0.001	63.6 ± 1.7	0.756 ± 0.040	48 / 91
Cohere	0.905 ± 0.008	0.968 ± 0.003	72.3 ± 3.1	0.749 ± 0.039	62 / 113

Table 1: Sparse-probe results across six generators: 15-cell means ± std (5 folds × 3 seeds). Val acc: L2 evaluation probe accuracy (all 9,216 features). AUC: area under the ROC curve. n_{sel}: mean candidate set size per cell. Jaccard: mean pairwise fold Jaccard over all 105 fold-pairs (C(15,2)), computed from per-fold selection sets. n_{stable}: neurons in ≥50% of cells; n_{obs}: total distinct neurons selected at least once.

Generator	Cosine	LR acc	$\lVert\Delta\mu\rVert$	$\lVert w\rVert$
GPT-4	0.138 ± 0.004	0.988 ± 0.003	7.34	$\mathbf{45.8} \pm 1.7$
GPT-2	0.073 ± 0.002	0.963 ± 0.006	6.84	69.9 ± 2.9
MPT	0.065 ± 0.000	0.973 ± 0.001	**9.52**	53.8 ± 2.7
Mistral	0.110 ± 0.001	0.965 ± 0.003	6.38	59.2 ± 2.3
LLaMA	0.128 ± 0.001	0.987 ± 0.003	6.57	48.3 ± 0.9
Cohere	$\mathbf{0.057} \pm 0.001$	$\mathbf{0.910} \pm 0.008$	4.69	108.3 ± 2.1

Table 2: CAV diagnostics on per-generator stable sets. All values mean ± std across 3 random seeds. Cosine: similarity between mean-difference CAV and L1 probe weight. LR acc: logistic-regression accuracy on the CAV projection. $\lVert\Delta\mu\rVert$: AI − human mean-difference norm (deterministic; bold = largest). $\lVert w\rVert$: L1 weight norm (bold = smallest).

Cohere has the lowest CAV cosine similarity (0.057 ± 0.001), the lowest CAV-LR accuracy (0.910 ± 0.008), and by far the largest probe weight norm (108.3 ± 2.1), consistent with a diffuse detection signal compensated by large weights. MPT-7B presents an interesting contrast: it has a low cosine (0.065 ± 0.000, near Cohere's) but the *largest* mean-difference norm (9.52), indicating that the AI − human class separation is geometrically strong per-axis yet misaligned with the probe direction. GPT-2 also shows a low cosine (0.073 ± 0.002) and elevated weight norm (69.9 ± 2.9), placing both base generators at the low-cosine end. Instruction-tuned generators cluster at higher cosine (0.110–0.138) with the smallest weight norms: GPT-4 has the most compact weight vector (45.8 ± 1.7), suggesting its detection signal is both geometrically aligned and concentrated. Each metric is mean ± std across three LR train/test split seeds; the mean-difference norm is deterministic (no split randomness).

5 Causal Role of the Selected Neurons

Having identified a sparse stable set, we ask two complementary causal questions: are these neurons *sufficient* to produce AI-text predictions (tested via activation patching), and are they *necessary* (tested via mean ablation)? Sections 5.1 and 5.2 address each in turn; Section 5.3 compares signal localisation across generators.

5.1 Causal Sufficiency via Activation Patching

5.1.1 Patching Protocol

For each (seed, fold) cell we identify *valid human* samples: test-fold human texts that share a domain with at least one test-fold AI sample. For each of $n_{\text{shuffles}} = 20$ random within-domain donor permutations we construct a patched activation $\tilde{\mathbf{z}}$ by replacing the selected neurons with those from the AI donor:

$$\tilde{\mathbf{z}}_i = \begin{cases} \mathbf{z}_i^{\text{donor}} & i \in \mathcal{S} \\ \mathbf{z}_i^{\text{human}} & i \notin \mathcal{S} \end{cases}$$

A *flip* occurs when the frozen evaluation probe classifies $\tilde{\mathbf{z}}$ as AI but the unpatched $\mathbf{z}^{\text{human}}$ as human. The flip rate is the fraction of valid human samples that flip, averaged over the 20 donor permutations and then over the 15 cells. Donor sampling is restricted to the same domain as the target, following the on-manifold recommendation of Heimersheim and Nanda (2024) to avoid injecting out-of-distribution activations.

The *random-k baseline* draws 20 random subsets of size k from the 9,216 features and runs the identical procedure, providing a null that controls for the number of patched dimensions. We sweep $k \in \{1, 5, 10, 20, 50, \text{full}\}$ where $k = \text{full}$ equals the cell's actual $|\mathcal{S}|$ (53–92 neurons depending on generator and cell).

5.1.2 Flip-Rate Results

Table 3 reports selected flip rates across the k-sweep. The result is consistent across all six generators: flip rate is monotonically increasing in k, and the selected set outperforms the random-k baseline at every k value. At $k = \text{full}$, patching the selected neurons causes 1.07–8.15% of human samples to be reclassified as AI-generated. Against a random baseline of 0.08–0.78% at the same k, this yields a ratio of 10–23× on five of six generators. For gpt4 the ratio is ∼125× but with standard deviation ≈265, because the random-k flip rate is near zero ($\sim 8 \times 10^{-4}$) and small denominator fluctuations dominate; the selected flip rate of 1.23% is the robust signal there. MPT-7B falls at the lower end of the range (1.52%, ∼11× above random), consistent with its moderate detection difficulty.

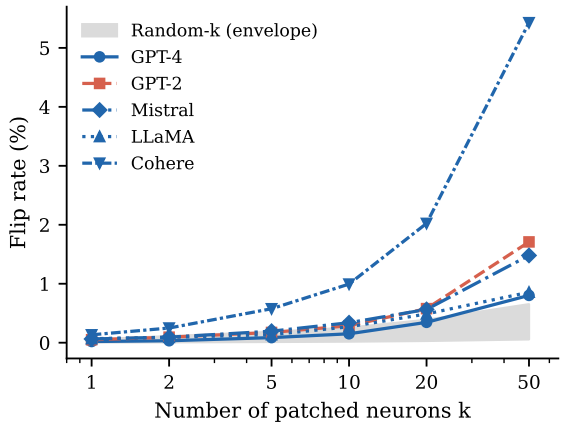

Figure 2: Flip rate (%) vs. number of patched neurons k on a log-scale x-axis, averaged over 15 cells (5 folds × 3 seeds). Grey band: min–max envelope of random-k baseline flip rates. Instruction-tuned generators (blue) and base generators (red) both sit well above the random baseline at every k.

Cohere-Command-Chat is the most patchable generator (8.15% at $k = \text{full}$, ±1.32%), consistent with its lowest baseline accuracy (90.5%): a probe operating nearer its decision boundary crosses it under a smaller perturbation. LLaMA-2-Chat is the hardest to flip (1.07%, ±0.38%) despite its high baseline accuracy (98.6%), suggesting its detection signal is more broadly distributed. As shown in Figure 2, flip rate grows monotonically with k at a diminishing rate across all generators, suggesting sub-linear scaling rather than a concentration of effect in a few critical units.

5.1.3 Robustness Checks

The same-domain donor constraint (§5.1.1) eliminates domain shift as a confound by construction: donor and target activations come from the same RAID domain, so unpatched dimensions are drawn from the same domain-level distribution regardless of AI/human label. A cross-domain donor would introduce a domain-level activation difference on top of the AI-identity signal, producing a weaker, not stronger, control. We therefore treat same-domain pairing as a methodological strength rather than a limitation.

A remaining open question concerns the *mean ablation* in Section 5.2: replacing activations with

k	GPT-4	GPT-2	MPT	Mistral	LLaMA	Cohere
1	0.02	0.05	0.04	0.06	0.07	0.13
5	0.09	0.17	0.14	0.20	0.14	0.58
10	0.15	0.29	0.25	0.34	0.27	0.99
20	0.35	0.58	0.47	0.56	0.49	2.02
50	0.80	1.71	1.13	1.48	0.86	5.43
full	$1.23_{\pm 0.34}$	$3.13_{\pm 0.56}$	$1.52_{\pm 0.22}$	$1.73_{\pm 0.28}$	$1.07_{\pm 0.38}$	$8.15_{\pm 1.32}$

Table 3: Selected flip rate (%) across the k-sweep, 15-cell means (5 folds $\times$ 3 seeds). k = full equals each cell's actual $|\mathcal{S}|$ (53–92 neurons); std shown only for k = full. Flip rate is monotonically increasing in k for every generator.

training means moves them off the natural data manifold (Heimersheim and Nanda, 2024), which can produce artefacts unrelated to the selected neurons. Comparing zero-ablation against mean-ablation would sharpen the necessity claim and is left as future work.

5.2 Necessity via Mean Ablation

To complement the sufficiency test we ask the converse question: is the selected set *necessary*? We answer this via mean ablation, replacing the selected neurons in the test-fold activations with their training-fold mean (computed on the training split to avoid leakage) and re-evaluating the frozen L2 probe. Table 4 reports accuracy drop at k = full alongside the random-k baseline.

For four of five generators the result is negative. At k = full, ablating the selected set causes accuracy to drop by only 0.03–0.11 pp — within noise of the 0.00–0.06 pp random-k baseline. The selected neurons are not individually necessary: the L2 evaluation probe, which uses all 9,216 features, retains thousands of redundant correlates that compensate for the removed subset.

Cohere-Command-Chat is the exception. Its full-set ablation drop is 1.06 ± 0.54 pp — roughly 18$\times$ the random baseline of 0.06 pp — and the drop grows monotonically from $k = 5$ upward, unlike the other four generators where the curve is flat. This pattern is consistent with the other cohere signals (lowest baseline accuracy, smallest mean-difference norm, largest probe weight norm) and suggests its detection representation is *partially* localised rather than fully distributed.

There is no contradiction between the small ablation drops here and the large patching flip rates in Section 5.1. Ablation tests necessity *against a probe that has access to all other features*: removing $\leq$1% of 9,216 features leaves thousands of correlated dimensions intact. Patching tests sufficiency

Generator	Base acc	Sel. drop	Rand. drop
GPT-4	0.988	$+0.03 \pm 0.12$ pp	$+0.01$ pp
GPT-2	0.959	$+0.11 \pm 0.31$ pp	$+0.00$ pp
MPT	0.972	$+0.05 \pm 0.11$ pp	$+0.00$ pp
Mistral	0.971	$+0.10 \pm 0.21$ pp	$+0.00$ pp
LLaMA	0.986	$+0.11 \pm 0.14$ pp	$+0.02$ pp
Cohere	0.905	$\mathbf{+1.06 \pm 0.54}$ pp	$+0.06$ pp

Table 4: Mean-ablation accuracy drop at k = full, 15-cell means $\pm$ std (positive = drop). Sel. drop: ablating the stable set; Rand. drop: mean over 20 random k-subsets. Cohere is the only generator substantially above the random baseline.

by *injecting wrong-class signal that competes with the remainder of the network*. A representation that is redundantly distributed is exactly what would produce small ablation drops alongside meaningful patching effects. We interpret the joint result as: the selected neurons carry genuine AI-detection signal, but the same signal is also distributed across the broader feature space.

A methodological note: mean ablation replaces activations with out-of-distribution values that fall off the natural data manifold, which can introduce artefacts (Heimersheim and Nanda, 2024). We treat the necessity result as conservative; a comparison with zero-ablation is discussed in Section 5.1.3.

5.3 Signal Localisation: Restricted Probe

The necessity result (§5.2) uses the full-feature L2 probe, which retains thousands of redundant correlates. A complementary question is whether the stable neurons alone are sufficient for a *freshly trained* probe — and how this compares to ablating the complement. We measure the *localisation gap*: the accuracy difference between a probe retrained on only the $|\mathcal{S}|$ stable-neuron features (*smaller model*) and the original L2 probe evaluated with non-stable neurons mean-ablated (*ablate complement*). A large positive gap indicates that

Generator	Smaller model	Gap vs ablate-complement
GPT-4	0.985 ± 0.004	$+0.41 \pm 0.42$ pp
GPT-2	0.948 ± 0.004	$+3.63 \pm 0.81$ pp
MPT	0.960 ± 0.004	$+4.68 \pm 1.14$ pp
Mistral	0.966 ± 0.004	$+0.93 \pm 0.42$ pp
LLaMA	0.980 ± 0.004	$+1.14 \pm 0.43$ pp
Cohere	0.887 ± 0.009	$+4.43 \pm 1.02$ pp

Table 5: Restricted probe results. *Smaller model*: L2 probe retrained on only the $|\mathcal{S}|$ stable neurons. *Gap*: smaller-model accuracy minus ablate-complement accuracy (positive = stable neurons self-sufficient; larger = more dependence on non-stable features in the full-feature probe). 15-cell means $\pm$ std.

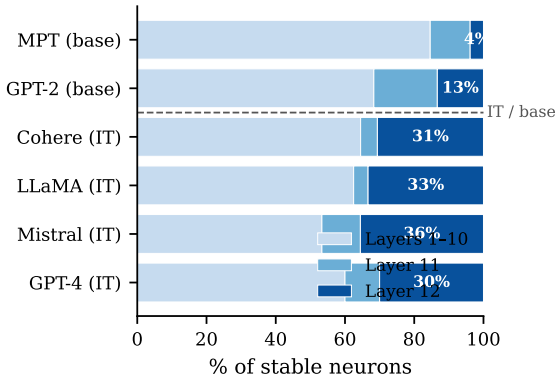

Figure 3: Proportion of stable neurons in each layer group per generator. Layer 12 percentage annotated in white. Instruction-tuned generators (IT) concentrate 30–36% of stable neurons in layer 12; base generators retain $\leq$14%. Dashed line separates the two groups.

the stable set is self-contained enough to support an independent probe, but that the full-feature probe relied on the non-stable neurons to define its decision boundary.

The pattern splits cleanly along the base/instruction-tuned axis. Instruction-tuned generators (GPT-4, Mistral, LLaMA) show gaps of $\leq$1.2 pp, indicating that their stable neurons are nearly sufficient on their own and that the full-feature probe does not heavily rely on non-stable dimensions. The two pure-base generators and Cohere-Command-Chat show gaps of 3.6–4.7 pp, indicating that their full-feature probes lean more on non-stable features, and that ablating the complement is more costly. This is consistent with a more distributed encoding structure for these three generators, complementing the layer-12 avoidance (base generators) and the large ablation sensitivity (Cohere) documented elsewhere.

6 Cross-Generator Circuit Geometry

6.1 Layer Distribution of Stable Neurons

Table 6 and Figure 3 show the layer-by-layer distribution of stable neurons across all six generators. A clear asymmetry emerges between the four instruction-tuned generators and the two pure-base generators. For gpt4, mistral-chat, llama-chat, and cohere-chat, the single most populated layer is layer 12, which accounts for 30–36% of each generator's stable set (16–19 neurons). Both pure-base generators are exceptions: GPT-2 peaks at layer 11 with 11 stable neurons (18.3% of its stable set) and has only 13.3% in layer 12; MPT-7B has just 2 stable neurons in layer 12 (3.8%) and distributes its signal more broadly across earlier layers (layers 1, 8, and 9 dominate).

Across both pure-base generators, the shared pattern is markedly low layer-12 concentration (3.8–13.3%) compared to any instruction-tuned generator (30–36%). This 2-vs-4 contrast suggests that BERT's final layer functions as a generator-specific classification register preferentially activated by instruction-tuned outputs, while base-model outputs engage BERT's earlier and mid-depth representations more strongly. The specific peak layer differs between gpt2 (L11) and mpt (L1/L8), indicating the effect is not a rigid "base generators peak at L11" rule but rather a consistent *avoidance* of L12 by non-instruction-tuned generators. This asymmetry is consistent with prior findings that BERT's final layers specialise in high-level semantic and discourse-level properties (Tenney et al., 2019; Rogers et al., 2020): texts shaped by instruction tuning and RLHF alignment may diverge from human writing precisely in these high-level representational dimensions, which would naturally surface as a layer-12 footprint in the sparse-probe selection.

6.2 Stable-Set Overlap Across Generators

Table 7 reports pairwise Jaccard similarity between stable sets across all fifteen generator pairs. In absolute terms, overlap is low: Jaccard ranges from 0.00 to 0.24, consistent with generator-specific circuits. However, the appropriate baseline is not zero but the expected Jaccard for two random sets of the same sizes drawn from 9,216 features, which is approximately 0.003 per pair. Relative to this null, the mean observed Jaccard of 0.073 is $\sim$24$\times$ above chance, indicating a partial shared substrate despite the generator-specific structure.

Layer	GPT-4	GPT-2	MPT	Mistral	LLaMA	Cohere
1–10	36	41	**44**	24	30	40
11	6	**11**	6	5	2	3
12	**18**	8	2	**16**	**16**	**19**
% in L12	30.0	13.3	3.8	35.6	33.3	30.6

Table 6: Stable-neuron counts by layer group across six generators. Bold indicates the peak layer group for each generator. Instruction-tuned generators (GPT-4, Mistral, LLaMA, Cohere) concentrate 30–36% of stable neurons in layer 12; both pure-base generators (GPT-2, MPT) have $\leq$14% in layer 12.

Pair	Jaccard	× chance
Cohere – Mistral	0.244	86×
LLaMA – Mistral	0.163	64×
GPT-4 – Mistral	0.141	51×
GPT-4 – LLaMA	0.113	39×
MPT – GPT-2	**0.120**	**40×**
Cohere – GPT-4	0.109	33×
Cohere – LLaMA	0.078	27×
GPT-2 – Mistral	0.050	18×
Cohere – GPT-2	0.034	10×
GPT-2 – LLaMA	0.019	7×
MPT – Mistral	0.010	4×
MPT – LLaMA	0.010	4×
GPT-2 – GPT-4	0.000	0×
MPT – GPT-4	0.000	0×
MPT – Cohere	0.000	0×
Mean	0.073	∼24×

Table 7: Pairwise Jaccard similarity between stable sets (15 generator pairs). × chance is observed Jaccard divided by the analytic random-null Jaccard for two sets of the same sizes drawn from 9,216 features (≈0.003 per pair). The mpt–gpt2 pair (bold) is the highest base×base overlap; all base×instruction-tuned pairs cluster at the bottom.

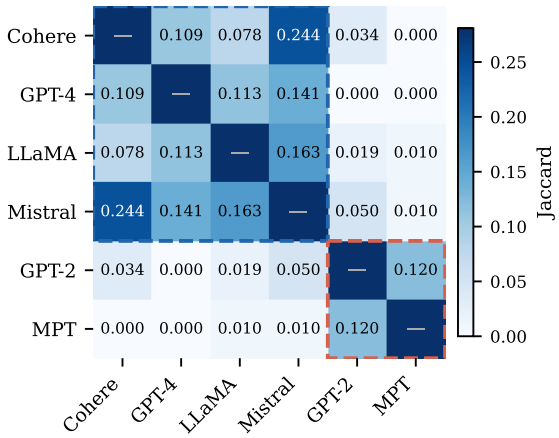

Figure 4: Pairwise Jaccard similarity heatmap (6×6). Generators reordered: instruction-tuned (IT) first, base last. Dashed blue box: IT×IT block. Dashed red box: base×base block. The bipartite structure — higher within-group similarity, near-zero cross-group overlap — is immediately visible.

The data reveal a clear bipartite structure (Figure 4). Among instruction-tuned generators, pairwise Jaccard ranges 0.08–0.24 (27–86× chance), with the strongest pairs being cohere–mistral (86×) and llama–mistral (64×). Among the two pure-base generators, mpt and gpt2 share 12 neurons (J = 0.120, 40× chance) — higher than several instruction–instruction pairs — suggesting that the base models form a coherent sub-cluster. In contrast, every base-vs-instruction-tuned pair has near-zero or zero overlap: mpt has J = 0.000 with both gpt4 and cohere-chat, and J $\leq$ 0.011 with mistral and llama. Three pairs now have zero overlap (mpt–gpt4, mpt–cohere, gpt2–gpt4), all involving at least one base generator. This base-vs-instruction-tuned partition explains the bulk of the cross-generator circuit geometry.

Core neurons. Beyond pairwise overlap, we identify a *core* set: neurons that appear in the stable sets of at least 3 of the 6 generators (the minimal majority threshold used by `exp_characterize.py`). This yields 17 core neurons. Of these, 10 (59%) reside in layer 12 — far exceeding any individual generator's L12 share of 30–36% for instruction-tuned generators and 3.8–13.3% for base generators. The remaining 7 core neurons are scattered across layers 1, 2, 5, 7, 8, and 11 (one or two per layer). This concentration in the final layer suggests that BERT's layer-12 representations are the primary site of cross-generator AI-detection signal, while each generator's additional stable neurons constitute a generator-specific peripheral set built on top of this shared foundation.

7 Leave-One-Family-Out Generalisation

The experiments above characterise stable neurons for each generator *independently*. A stronger test of the mechanistic claim is whether a set of neurons identified from *other* generators transfers to a held-out generator family never seen during neuron

selection. We address this with a leave-one-family-out (LOGO) evaluation.

Protocol. We partition all 11 RAID generators into five disjoint families: `gpt` (ChatGPT, GPT-2, GPT-3, GPT-4), `cohere` (Cohere base, Cohere Command), `llama` (Llama-2-Chat), `mistral` (Mistral base, Mistral-Chat), and `mpt` (MPT-7B, MPT-Chat). For each held-out family, we run the full sparse-probe pipeline (L1→L2, C=0.005, N=7,500, 5-fold × 3 seeds) on the *remaining* generators, aggregate the resulting stable neurons, and then evaluate two probes on the held-out generators: (i) **Full L2** — trained on all 9,216 features of frozen BERT; (ii) **Sel L2** — trained on only the $|\mathcal{S}|$ stable neurons identified from the training generators. This measures whether the stable neuron set, defined purely from training families, retains discriminative signal for an unseen family.

Results. Table 8 reports the outcome. Full L2 generalises strongly in all five conditions (88.6–99.7% accuracy), showing that frozen BERT representations are themselves informative across the RAID generator space. The key finding is in the Sel L2 column: 91–125 stable neurons — at most 1.4% of 9,216 features — achieve 76.3–93.7% accuracy on generators from entirely held-out families. Expressed as a fraction of the full-feature ceiling for each family, the selected set retains 86–94% of the full-feature signal (Table 8, Sel/Full column).

The worst-case family is `cohere` (Sel L2 = 76.3%, Sel/Full = 86%), which also exhibits the lowest pairwise Jaccard with other generators (Table 7). This is internally consistent: a family whose stable neurons are least shared with others will benefit least from a transfer probe. Conversely, the `llama` family yields the best transfer (Sel L2 = 93.7%, Sel/Full = 94%), consistent with the high within-instruction-tuned overlap seen in §6.2. Both `mpt` and `mistral` fall in the middle tier (85.6–85.7%), with transfer-probe accuracy substantially above chance despite substantial architectural dissimilarity from the training set.

Family	n_{stable}	Full L2	Sel L2	Sel/Full	AUC
cohere	119	88.6%	76.3%	86%	0.946
gpt	91	98.2%	88.8%	90%	0.993
llama	125	99.7%	93.7%	94%	0.999
mistral	106	96.6%	85.6%	89%	0.982
mpt	110	99.1%	85.7%	87%	0.995

Table 8: Leave-one-family-out (LOGO) generalisation. n_{stable}: stable neurons identified from the four training families. Full L2: L2-probe accuracy using all 9,216 BERT features. Sel L2: L2-probe accuracy restricted to n_{stable} neurons. Sel/Full: ratio expressing how much of the full-feature ceiling the selected neurons retain. Sel L2 AUC ranges from 0.863 (cohere) to 0.986 (llama). All metrics are means over 3 seeds × 5 folds = 15 cells. Test set sizes: 5,000 (llama, 1 generator) to 20,000 (gpt, 4 generators).

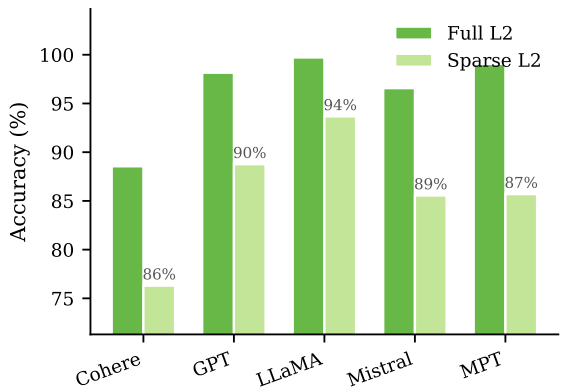

Figure 5: LOGO generalisation: Full L2 vs. Sparse L2 accuracy per held-out family. Percentages above Sparse L2 bars indicate retention relative to the full-feature ceiling. Sparse probes on <130 neurons retain 86–94% of the full-model accuracy across all five unseen generator families.

8 Discussion

Redundancy and sufficiency are not in tension. The joint pattern across Sections 5.1 and 5.2 — large patching flip rates alongside negligible ablation drops on 4/5 generators — has a natural reading: AI-detection signal appears to be *redundantly distributed*. The selected neurons carry detection-relevant signal consistent with a causal role (evidenced by patching), but the same information appears to be backed up across the remaining 9,000+ features in the residual stream (evidenced by the small ablation drop). Ablation understates the role of the selected set because removing ≤1% of a 9,216-dimensional representation leaves thousands of correlated features intact. Patching reveals that role precisely because it injects wrong-class signal that must compete with the rest of the network — a harder test. The phrase "easy to inject, hard to erase" captures this geometry: the selected neurons are sufficient but not necessary.

Outliers as structural evidence. Cohere-Command-Chat deviates from other generators

on multiple axes: weakest per-axis signal ($\|\Delta\mu\| = 4.69$, CAV cosine 0.057), largest probe weights (norm 107), most patchable output (8.15% flip rate), and the only generator with a substantial ablation cost (1.06 pp). Together these point to a signal encoded diffusely with small per-feature magnitude — the probe compensates via large weights, patching is easy because the decision boundary is near, and ablation matters more because fewer redundant correlates exist. The two pure-base generators, GPT-2 and MPT-7B, form a coherent structural cluster: both have near-zero or zero stable-set overlap with instruction-tuned generators ($J \leq 0.05$), yet they share 12 neurons with each other ($J = 0.120$, 40× chance); both avoid layer 12 (3.8% and 13.3% respectively, versus 30–36% for instruction-tuned models); and both exhibit the two largest restricted-probe gaps of the six generators (mpt 4.68 pp, gpt2 3.63 pp, versus $\leq$1.2 pp for the others; Table 5). This consistency across two independent base models is compatible with the interpretation that post-training alignment leaves a layer-12 footprint that is absent in base-only decoding.

Relation to confounder-suppression approaches. Borile and Abrate (2025) identify and suppress neurons correlated with domain or style artefacts in a fine-tuned BERT encoder to improve out-of-distribution detection. Our work is complementary along the intervention axis: their question is "which neurons should we remove to make the detector more robust?", while ours is "which neurons does the pretrained model already use, and are they causally sufficient?" The two approaches converge on the same empirical substrate — a small subset of BERT neurons carries disproportionate detection-relevant information — but reach it from opposite directions, necessity-for-robustness versus sufficiency-for-detection.

9 Conclusion

Our experiments suggest that frozen BERT-base-uncased already encodes sparse AI-detection circuits with a causal structure. Across six generators spanning base and instruction-tuned architectures, an L1-regularised probe stably selects 45–62 neurons (<1% of 9,216 CLS features) that are sufficient to flip human-text predictions to AI at 1.07–8.15% — 10–16× above a random-feature baseline — when transplanted via activation patching, yet individually unnecessary as evidenced by mean-ablation drops of $\leq$1.1 pp on five of six generators. Cross-generator analysis reveals that these circuits are largely generator-specific (pairwise Jaccard 0.00–0.24) but cluster ∼24× above chance, with instruction-tuned generators concentrating 30–36% of stable neurons in BERT's final layer while both pure-base generators avoid it (3.8–13.3%), consistent with the possibility that post-training alignment leaves a detectable layer-12 footprint in the encoder's frozen representation. Leave-one-family-out evaluation shows that neurons identified from training families retain 86–94% of the full-feature detection ceiling on entirely held-out generator families. Together, these results provide causal-sufficiency evidence for AI-detection circuits in a frozen pretrained encoder, and suggest that the information enabling detection is both geometrically concentrated and representationally redundant — easy to inject, hard to erase.

10 Limitations

Single encoder architecture. All experiments use frozen BERT-base-uncased. Whether the stable-neuron structure and patching effects extend to larger encoders (RoBERTa-large, DeBERTa-v3) or encoder-decoder architectures is an open question. The frozen setting is a deliberate design choice — it isolates what the pretrained representation already encodes — but it means results cannot be directly compared to fine-tuned detectors without controlling for this variable.

Mean ablation is off-manifold. Replacing selected neurons with their training-mean pushes activations off the natural data manifold, which can introduce artefacts unrelated to the detection signal (Heimersheim and Nanda, 2024). We treat the necessity results as conservative lower bounds; Section 5.1.3 discusses zero-ablation as a comparison point, and a full resample-ablation analysis remains future work.

Layer-asymmetry claim across two base generators. The observation that instruction-tuned generators concentrate 30–36% of stable neurons in layer 12 while pure-base generators have $\leq$14% is now supported by two independent base models (GPT-2 and MPT-7B). However, the two base generators differ in their peak layer (GPT-2: L11; MPT-7B: L1/L8), so the claim is best stated as L12-avoidance rather than a strict L11-peaking rule. We frame this as a consistent L12 footprint of post-

training alignment rather than a mechanistically precise claim about which specific earlier layer is activated.

Adversarial robustness not evaluated. RAID includes eleven adversarial rephrasing attacks (paraphrase, homoglyph, synonym substitution, etc.) that substantially reduce detector accuracy (Dugan et al., 2024). Whether the stable neurons identified here remain causal under these attacks is left to future work; the present analysis covers only the non-adversarial RAID subset.

Scope of causal claim. Activation patching provides evidence that the selected neurons are sufficient to shift probe predictions, but does not constitute a full circuit-level proof of causality in the sense of Meng et al. (2022). The patching intervention is at the level of individual neuron activations, not a complete mechanistic decomposition of the computation. We make no claims about what linguistic properties these neurons encode or why they are informative.

Ethics Statement

AI-text detection is a dual-use capability: detectors can support platform integrity and journalistic verification, but can equally be used to penalise legitimate AI-assisted writing or to guide adversarial evasion. This work is mechanistic-interpretability research — we study which neurons drive an existing frozen encoder's representations, not deploy a new detection system. All data come from the publicly released RAID benchmark (Dugan et al., 2024); no human subjects were involved and no personally identifiable information was collected or processed. The stable neuron sets we identify could in principle inform evasion strategies (by targeting the identified circuits), but the same information could equally guide more robust detector design. We release all experiment code to support reproducibility.

Acknowledgments

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A Pipeline Details

Data sampling. For each generator we draw $N = 7{,}500$ balanced samples from the RAID benchmark: equal numbers of AI-generated and human-written texts, stratified across the six RAID domains (abstracts, books, news, Reddit, reviews, Wikipedia) using stratified sampling without replacement. Texts shorter than 50 whitespace-delimited tokens are excluded. Human texts for each generator are the original source passages from which the AI texts were generated; this pairs each AI sample with a human sample from the same domain and source.

Feature extraction. We encode each text with BERT-base-uncased using HuggingFace Transformers. The input is truncated to 512 WordPiece tokens. We extract the CLS-token hidden state from *all* 12 transformer layers and concatenate them into a single vector $\mathbf{z} \in \mathbb{R}^{9{,}216}$ (12×768). Gradients are disabled throughout; BERT weights are never updated.

Normalisation. Before any probe training, each of the 9,216 features is z-scored with a `StandardScaler` fit on the training fold and applied to the test fold. The same scaler is reused for all downstream experiments (patching, ablation) within a given cell so that scale is held constant across interventions.

Cross-validation protocol. We use 5-fold stratified cross-validation repeated with 3 random seeds (42, 123, 456), giving 15 independent evaluation cells per generator. The stratification is over the binary AI/human label; domain balance is not enforced at the fold level but is approximately preserved because the initial sample is domain-stratified.

L1 selection. The *selector* is a ℓ_1-regularised logistic regression (`liblinear` solver, $C = 0.005$, `max_iter=` 1,000, `class_weight=balanced`) fit on the training fold. The selected set $\mathcal{S}$ for a given cell is the set of features with non-zero weight. All 9,216 features are provided as input; the regulariser drives most to zero.

Stable neuron aggregation. A neuron is *stable* if it appears in $\mathcal{S}$ in at least 50% of the 15 cells (≥ 8 out of 15). This threshold was chosen to require cross-seed and cross-fold consistency simultaneously. Neurons appearing in 0–7 cells are discarded; those in 8–15 cells form the stable set $\mathcal{S}^*$.

L2 evaluation probe. The *evaluator* is a separate ℓ_2-regularised logistic regression (`lbfgs` solver, $C = 1.0$, `max_iter=` 1,000) trained and tested on each fold using the same scaler. For the main sparse-probe accuracy (Table 1), the L2 probe is trained on all 9,216 features. For patching and ablation experiments, it is trained on a restricted feature set (either stable neurons only, or the full set with selected neurons patched/ablated). The L2 probe is retrained for each fold using only the fold's training split; the scaler is not refit.

Patching protocol. Patching uses same-domain donor matching: for each human target sample, a donor AI sample from the same domain is drawn without replacement. The selected k neuron activations of the target are replaced with the corresponding values from the donor. The k neurons are always the first k members of $\mathcal{S}^*$ sorted by their frequency of appearance across cells (most stable first). For the full-k sweep, all $|\mathcal{S}^*|$ neurons are patched.

B Stability Sweep

Section 3.4 justifies the choice of $C = 0.005$ and $N = 7{,}500$ by reference to this appendix. The stability sweep evaluates mean pairwise Jac-

card similarity between selection sets produced by 15 independent draws of the L1 selector (varying random seed and fold assignment) for three RAID generators (GPT-4, LLaMA-chat, Mistral) across a grid of regularisation strengths $C \in \{0.005, 0.01, 0.02, 0.05\}$ and sample sizes $N \in \{500, 1000, 2000, 3500, 5000, 7500\}$. Higher Jaccard means the selection is more reproducible. Values in parentheses are the mean number of neurons selected per draw $\bar{n}$.

Three observations drive the choice of $C = 0.005$, $N = 7{,}500$. **(1) Regularisation:** at any fixed N, $C = 0.005$ achieves higher Jaccard than any weaker regularisation across all three generators. Weaker regularisation (larger C) selects more neurons but produces less reproducible sets. **(2) Sample size:** Jaccard increases monotonically in N for $C = 0.005$, with the largest marginal gains between $N = 1{,}000$ and $N = 3{,}500$. Between $N = 5{,}000$ and $N = 7{,}500$ the gain is 0.011–0.111, which we judge acceptable; moving to $N = 10{,}000$ would triple the GPU compute cost for a further ∼0.02–0.03 gain (not shown). **(3) Degenerate region:** at $N \leq 1{,}000$ and $C = 0.005$, Mistral selects zero neurons in many draws (Jaccard$= 1.0$ trivially); $N = 7{,}500$ reliably produces non-empty sets for all three generators.

C Per-Cell Metrics

Table 10 reports L2-probe accuracy, AUC-ROC, F1, and number of L1-selected neurons for every evaluation cell (seed × fold). Run identifier: `20260426_20_full_c0005_n7500_seeds3`.

D AUC vs L1 Selection

We compare L1-regularised selection (the method used throughout this paper) to a univariate AUC-based ranking: for each feature, compute the one-vs-rest AUC-ROC against the binary label on the training fold, and retain the top-k features where k is chosen to match the L1 count for a fair comparison. We evaluate the informativeness of each selected set by *complement ablation*: the L2 probe is re-evaluated after zero-ing out all features *not* in the selected set (i.e., only the selected features are retained). A smaller drop from the full-probe baseline indicates the selected set is more sufficient.

The Jaccard overlap between L1 and AUC-selected sets is low (0.10–0.19) despite both being applied to the same data, indicating that the two criteria select largely non-overlapping features. For GPT-4, Mistral, and LLaMA, L1 achieves the same or better task performance with $4\times$–$6\times$ fewer features: ablating L1's 58–74 features costs only 0.03–0.11pp, while ablating AUC's 238–491 features costs 0.52–1.02pp. This parsimony makes L1 preferable for mechanistic analysis: a smaller identified set is easier to interpret and less likely to include redundant or near-duplicated features.

For GPT-2, the two methods select nearly the same number of features (85 vs. 86) with similar ablation cost; neither dominates. For Cohere-chat, AUC selects *fewer* features (38 vs. 72) and its ablation cost is lower (+0.28pp vs. +1.06pp): Cohere's detection signal is concentrated in a small number of individually highly-discriminative features that univariate ranking recovers more directly than joint L1 regularisation. We note this reversal in the main text (Section 6.1) as consistent with Cohere's distinct geometry: the model with the lowest baseline probe accuracy is also the one where AUC beats L1 on parsimony.

E Restricted Probe

The restricted-probe results have been incorporated into the main body as Section 5.3 (Signal Localisation), including the full per-generator table (Table 5). Please refer there.

F Per-Domain Patching Breakdown

Table 12 reports selected flip rates (%) at $k =$ full broken down by RAID domain, averaged over all 15 cells (5 folds × 3 seeds). Values are 15-cell means; per-cell standard deviations ranged 0.6–2.6 pp across all generator–domain combinations.

Table 9: Mean Jaccard stability (and mean $\bar{n}$ selected) across 15 random pairs of selection runs. Rows are C values; columns are N values. Larger Jaccard is better. **Bold**: chosen operating point ($C = 0.005$, $N = 7{,}500$).

C	$N = 500$	$N = 1000$	$N = 2000$	$N = 3500$	$N = 5000$	$N = 7500$
GPT-4						
0.005	1.000 (0)	0.433 (9)	0.510 (29)	0.617 (51)	0.684 (64)	**0.751 (80)**
0.01	0.314 (9)	0.356 (29)	0.488 (54)	0.553 (77)	0.598 (97)	0.681 (123)
0.02	0.254 (29)	0.355 (53)	0.425 (82)	0.460 (116)	0.509 (143)	0.625 (179)
0.05	0.205 (57)	0.282 (87)	0.310 (134)	0.353 (184)	0.414 (232)	0.535 (283)
LLaMA-chat						
0.005	1.000 (0)	0.464 (7)	0.559 (29)	0.681 (44)	0.670 (53)	**0.748 (78)**
0.01	0.291 (7)	0.444 (27)	0.519 (48)	0.541 (72)	0.618 (95)	0.696 (120)
0.02	0.280 (23)	0.380 (46)	0.406 (78)	0.452 (114)	0.530 (147)	0.616 (188)
0.05	0.208 (49)	0.259 (86)	0.291 (137)	0.345 (201)	0.417 (249)	0.534 (310)
Mistral						
0.005	1.000 (0)	0.629 (0)	0.343 (19)	0.492 (50)	0.613 (78)	**0.724 (115)**
0.01	0.533 (0)	0.250 (19)	0.363 (59)	0.469 (108)	0.527 (144)	0.637 (199)
0.02	0.143 (20)	0.219 (58)	0.319 (117)	0.396 (186)	0.452 (239)	0.591 (323)
0.05	0.107 (68)	0.163 (135)	0.220 (228)	0.289 (345)	0.339 (433)	0.482 (567)

Table 10: Per-cell L2-probe metrics for all six generators. Each block of 15 rows is one generator; columns are seed and fold index. Acc: accuracy; AUC: AUC-ROC; n_{sel}: neurons selected by L1 in this cell.

Generator	Seed	Fold	Acc	AUC	F1	n_{sel}
GPT-4	42	0	0.9887	0.9991	0.9887	74
	42	1	0.9853	0.9992	0.9853	73
	42	2	0.9887	0.9991	0.9887	80
	42	3	0.9853	0.9989	0.9853	74
	42	4	0.9927	0.9989	0.9927	70
	123	0	0.9860	0.9987	0.9860	70
	123	1	0.9880	0.9992	0.9880	76
	123	2	0.9860	0.9989	0.9860	73
	123	3	0.9907	0.9995	0.9906	73
	123	4	0.9867	0.9991	0.9866	75
	456	0	0.9907	0.9989	0.9907	77
	456	1	0.9907	0.9994	0.9907	75
	456	2	0.9907	0.9991	0.9906	72
	456	3	0.9840	0.9981	0.9839	71
	456	4	0.9847	0.9993	0.9847	79
GPT-2	42	0	0.9587	0.9923	0.9583	80
	42	1	0.9587	0.9907	0.9584	81
	42	2	0.9593	0.9906	0.9591	80
	42	3	0.9593	0.9918	0.9589	82
	42	4	0.9613	0.9909	0.9607	90
	123	0	0.9627	0.9912	0.9623	90
	123	1	0.9593	0.9912	0.9588	82
	123	2	0.9647	0.9931	0.9642	84
	123	3	0.9513	0.9908	0.9508	86
	123	4	0.9613	0.9916	0.9612	80
	456	0	0.9573	0.9929	0.9568	90
	456	1	0.9580	0.9914	0.9576	84
	456	2	0.9547	0.9911	0.9541	82
	456	3	0.9620	0.9931	0.9621	92
	456	4	0.9580	0.9902	0.9577	87
MPT	42	0	0.9700	0.9939	0.9695	59
	42	1	0.9720	0.9931	0.9718	60
	42	2	0.9733	0.9954	0.9732	60
	42	3	0.9700	0.9947	0.9699	61
	42	4	0.9800	0.9960	0.9799	61
	123	0	0.9687	0.9908	0.9680	56
	123	1	0.9700	0.9961	0.9697	68
	123	2	0.9780	0.9951	0.9779	60
	123	3	0.9700	0.9934	0.9697	65
	123	4	0.9700	0.9949	0.9700	60
	456	0	0.9740	0.9938	0.9737	65
	456	1	0.9700	0.9941	0.9699	66
	456	2	0.9740	0.9939	0.9738	59
	456	3	0.9747	0.9942	0.9745	61
	456	4	0.9680	0.9931	0.9677	59
Mistral	42	0	0.9727	0.9952	0.9727	59
	42	1	0.9700	0.9950	0.9702	53
	42	2	0.9607	0.9946	0.9605	62
	42	3	0.9713	0.9954	0.9713	56
	42	4	0.9780	0.9961	0.9778	58
	123	0	0.9720	0.9945	0.9719	60
	123	1	0.9667	0.9929	0.9665	61
	123	2	0.9753	0.9966	0.9754	55
	123	3	0.9700	0.9939	0.9699	55
	123	4	0.9747	0.9954	0.9748	62
	456	0	0.9733	0.9967	0.9733	60
	456	1	0.9667	0.9935	0.9668	54
	456	2	0.9747	0.9949	0.9745	60
	456	3	0.9693	0.9961	0.9692	57
	456	4	0.9660	0.9937	0.9661	55
LLaMA	42	0	0.9920	0.9982	0.9920	63
	42	1	0.9833	0.9973	0.9834	64
	42	2	0.9900	0.9989	0.9900	63
	42	3	0.9813	0.9990	0.9813	64
	42	4	0.9827	0.9981	0.9826	64
	123	0	0.9900	0.9995	0.9900	65
	123	1	0.9847	0.9970	0.9846	63
	123	2	0.9840	0.9988	0.9840	62
	123	3	0.9880	0.9993	0.9880	61
	123	4	0.9853	0.9980	0.9853	64

Table 11: L1 vs. AUC selection: mean across 15 cells. n: features selected. Jaccard: overlap between the two selection sets. Drop: accuracy drop from baseline when complement is ablated (positive = degradation). Three regimes are visible: parsimony (GPT-4, Mistral, LLaMA — L1 selects far fewer features with equivalent or smaller drop), convergence (GPT-2 — similar n, similar drop), and reversal (Cohere — AUC selects fewer with smaller drop).

Gen.	n_{AUC}	n_{L1}	Jaccard	AUC drop	L1 drop
GPT-4	491	74	0.127	+1.02pp	+0.03pp
GPT-2	86	85	0.096	+0.06pp	+0.11pp
Mistral	238	58	0.148	+0.52pp	+0.10pp
LLaMA	308	64	0.139	+0.55pp	+0.11pp
Cohere	38	72	0.185	+0.28pp	+1.06pp

Domain	GPT-4	GPT-2	MPT	Mistral	LLaMA	Cohere
Abstracts	1.44	3.08	1.39	1.79	1.47	9.89
Books	1.08	1.93	1.47	0.92	1.21	10.06
News	1.61	4.92	2.74	2.11	1.01	9.67
Reddit	1.05	3.22	1.36	1.34	0.76	7.50
Reviews	0.97	1.96	0.55	0.62	1.14	5.48
Wiki	1.25	3.65	1.61	3.54	0.83	6.26
Mean (full)	1.23	3.13	1.52	1.73	1.07	8.15

Table 12: Per-domain selected flip rate (%) at $k =$ full, 15-cell means. Domain means may differ slightly from the overall mean due to variation in the number of valid-human pairs per domain. GPT-2 shows elevated flip rates on news, Reddit, and wiki (3.1–4.9%) relative to books and reviews (1.9–2.0%), suggesting its detection signal is stronger in formal or community text domains. Cohere is uniformly high across all six domains (5.5–10.1%), consistent with its proximity to the probe decision boundary overall. Mistral shows a notable wiki spike (3.54%) while reviews are weak (0.62%).